

PRATHMESH UPADHYAY

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PROFESSIONAL SUMMARY

Full Stack Engineer with production experience architecting high-throughput booking platforms, distributed task queues, and transactional financial systems. Proven expertise in Node.js, Next.js, Python, Redis distributed locking, and event-driven microservices. Experienced in designing low-latency architectures with verified performance benchmarks and applied AI/RAG integrations.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript (ES6+), TypeScript, SQL, HTML5/CSS3

Backend & Distributed Systems: Node.js, Express.js, FastAPI, Redis, Distributed Locking, Message Queues (BullMQ), WebSockets, System Design, REST APIs, Microservices

Databases & Storage: PostgreSQL, MongoDB, ChromaDB (Vector DB), Redis, Write-Ahead Logging (WAL)

AI & LLM Systems: OpenAI API, Google Gemini API, LangChain, LangGraph, RAG, Vector Embeddings, Prompt Engineering

DevOps & Tools: Git, GitHub, Docker, Postman, Linux, k6 (Load Testing)

PROFESSIONAL EXPERIENCE

Urban Culture — Full Stack Developer

May 2026 – Present | Jaipur, India

- **Distributed Concurrency & Locking:** Engineered concurrent booking and slot-hold microservices using Redis SETNX distributed locking with automated TTL expiration, eliminating double-booking race conditions under 100+ simultaneous reservation attempts.
- **Checkout Recommendation Engine:** Architected personalized recommendation pipeline evaluating Apriori rule mining against ML re-ranking; conducted bootstrapped CI testing to fix data leakage, driving an 18% lift in checkout add-on conversions across 10,000+ monthly reservations.
- **Double-Entry Financial Ledger:** Designed high-integrity transactional wallet and ledger system in Node.js and Firestore; enforced atomic state transitions and audit logging, achieving 100% balance reconciliation accuracy across user balances, refunds, and partner payouts.
- **In-House Notification & Scheduling Engine:** Architected an event-driven notification platform utilizing Pub/Sub event streaming and a priority preemption algorithm to dynamically pause, restore, and re-queue user journeys; unified BigQuery OLAP with MongoDB via an LLM query layer and distributed task scheduler, driving an incremental +5 bookings/day.

Mentaive — Full Stack Engineering Intern

Feb 2026 – May 2026 | Remote

- **Platform Engineering & Scheduling:** Built production assessment platform using MERN stack and Next.js; developed automated cron job scheduling engine for time-bounded examination windows and automated alerts via SendGrid, cutting manual administrative overhead by 40%.
- **API Contracts & Cross-Stack Integration:** Coordinated 10+ RESTful API contracts between Next.js frontend and backend services, implementing strict request validation schemas that reduced cross-team sprint integration friction by 50%.
- **Frontend Performance Optimization:** Profiled component render lifecycles with React DevTools; eliminated recursive DOM re-renders and implemented route-based dynamic code splitting, improving Core Web Vitals and boosting page load speeds by 30% on low-bandwidth devices.

KEY ENGINEERING PROJECTS

TaskFlow — Distributed Task Queue Engine | Node.js, Redis, Express, WebSockets, k6, Lua

Mar 2025 – May 2025

- **Cluster Hash-Slot Architecture:** Architected a distributed background job engine utilizing Redis Cluster hash tags {queue} to enforce hash-slot locality across 16,384 shards, completely eliminating CROSSSLOT partition failures during atomic multi-key transactions.
- **Atomic Lua Dequeuing & Orphan Reaper:** Authored atomic Lua scripts for deterministic single-roundtrip dequeue operations (RPOP + ZADD); engineered an autonomous Orphan Reaper daemon with heartbeat monitoring and full-jitter exponential backoff retries.
- **Benchmarking & Resilience:** Stress-tested under 16 concurrent workers processing 100,000 background jobs via k6; sustained 3,351 jobs/sec throughput with 0% dropped tasks, sub-9.8ms P99 latency, and 100% crash recovery under injected SIGKILL.

GrabASeat — Real-Time Booking Engine | Next.js 14, Node.js, Express, Socket.io, MongoDB, Python, k6

Nov 2024 – Feb 2025

- **Real-Time Concurrency Control:** Built interactive seat reservation engine utilizing Socket.io event broadcasting and Redis distributed locking (SETNX with 10-minute TTL), guaranteeing mutual exclusion across 50+ simultaneous checkout requests.
- **ML Dynamic Pricing Microservice:** Developed isolated Python/Flask microservice running Random Forest regression; load-tested to 500 concurrent virtual users via k6 with P99 latency <120ms; reduced MongoDB query latency by 60% via compound indexing.
- **Secure Payment & Automated Ticketing:** Integrated Razorpay payment gateway with automated digital ticket generation using dynamic QR code verification.

ChronoStore — Time-Series Storage Engine | Node.js, Express, Google Gemini API, ChromaDB

Jun 2025 – Aug 2025

- **Append-Only Storage & WAL Engine:** Built custom append-only log file storage engine with in-memory range indexes, delivering sub-5ms query response times over 1,000,000+ data points; implemented Write-Ahead Logging (WAL) for fault-tolerant crash durability.
- **Semantic RAG Interface:** Designed semantic NL-to-query layer using Google Gemini function calling; embedded document context into ChromaDB vector database with hybrid search to synthesize complex analytical queries.
- **Data Compression:** Applied delta and run-length compression algorithms on sequential time-series timestamps and metric payloads, reducing on-disk storage footprint by 70%.

EDUCATION

Vellore Institute of Technology (VIT)

Vellore, India

Bachelor of Technology in Computer Science and Engineering

2022 – 2026

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Distributed Systems, Object-Oriented Programming